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### Q1: Complete Questions:

- 1) The primary object used to create a DataFrame in PySpark is the `SparkSession`
  - 2) The RDD Action used to return only the first element of an RDD is `first()`
  - 3) The RDD transformation used to split each element in the RDD into multiple new elements, often used after reading a text file, is `flatMap()`
  - 4) In PySpark DataFrames, the transformation function used to rename a column without dropping the old one is `withColumnRenamed()`
  - 5) When defining a manual schema, the structure used to represent a single column and its properties (name, type, nullability) is the `StructField` object.
  - 6) The `pyspark.sql.functions` function used to explicitly specify an ascending sort direction is `asc()`
  - 7) The RDD transformation used to create a new RDD where each original element is passed through a function to produce a new element is `map()`
  - 8) The DataFrame Action used to print a number of rows nicely to the console is `show()`
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### Q2: Written Questions:

#### 1. Difference between Transformation and Action:

- **Transformation:** defines a data processing step but does **not execute immediately** (`map()`, `filter()`,.....).
- **Action:** triggers execution and **returns a result** or outputs data (`count()` , `show()`,.....).

#### 2. Alternative function to `sort()`:

- `orderBy()`
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### Q3: Code Questions:

#### 1. Data Selection and Aliasing:

```
from pyspark.sql.functions import col
```

```
spark_df.select(col("ORIGIN_COUNTRY_NAME").alias("Source_Country")).show(2)
```

## 2. Creating DataFrame with Manual Schema:

```
from pyspark.sql import Row
from pyspark.sql.types import StructType, StructField, StringType, IntegerType

myManualSchema = StructType([
    StructField("name", StringType(), True),
    StructField("age", IntegerType(), True)])

myRowData = [Row("Alice", 30), Row("Bob", 25)]

myDf = spark.createDataFrame(myRowData, schema=myManualSchema)
myDf.show()
```